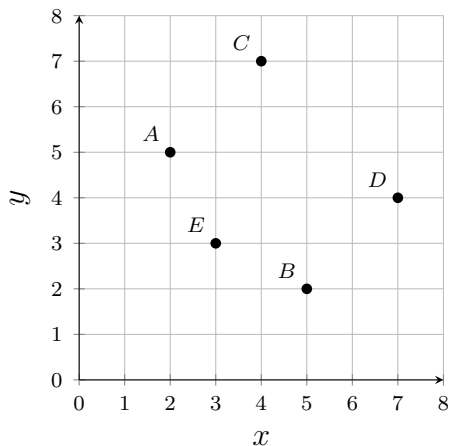


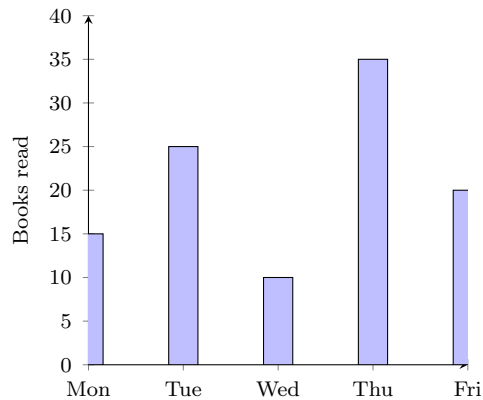
# Finding Order and Scale Mistakes on Graphs

## Directions

- Two displays are shown below. Every problem uses one of them.
- Two traps run through this sheet: writing an ordered pair back to front, and counting marks on an axis as if each one were worth 1.
- Answer the number part first, then write the explanation in full sentences.



**Display 1 — The Grid.** Both axes are marked every 1 unit.



**Display 2 — The Book Graph.** The vertical axis is marked every 5 books.

1. Ravi was asked to plot the point  $(2, 5)$  on the Grid. His mark landed on point  $B$  instead.

(a) Write the ordered pair for point  $B$ : ( \_\_\_\_\_ , \_\_\_\_\_ )

(b) Explain the mistake Ravi made.

**2.** Use the Book Graph. Leo says Thursday shows 7 books, because the bar reaches the 7th mark above zero. How many books does Thursday actually show?

Answer: \_\_\_\_\_

**3.** Use the Book Graph. Sam counted the marks on the Wednesday bar and reported 2 books.

(a) How many books does Wednesday actually show?

Answer: \_\_\_\_\_

(b) Explain Sam's mistake.

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**4.** On the Grid, points  $C$  and  $D$  are a reversed pair: one of them is  $(4, 7)$  and the other is  $(7, 4)$ . Measured left to right, how far apart are they?

Answer: \_\_\_\_\_ units

**5.** Use the Book Graph. Mia counted the marks on each bar and added those counts, reporting 21 books for the week. How many books were actually read that week?

Answer: \_\_\_\_\_

**6.** The same book data is redrawn on a new graph whose vertical axis is marked every 2 books instead of every 5. A student looks at the new Friday bar and says “it is taller, so more books were read on Friday”.

(a) On the new axis, how many marks tall is the Friday bar?

Answer: \_\_\_\_\_

(b) Were more books read on Friday? Explain.

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**7.** Use the Book Graph. A student found the mean by averaging the mark counts and reported 4.2 books per day.

(a) Write the actual mean number of books per day.

Answer: \_\_\_\_\_

(b) Write the range of the five daily totals.

Answer: \_\_\_\_\_

**8.** A student is told to plot  $(3, 30)$  on a graph whose horizontal axis is marked every 1 unit and whose vertical axis is marked every 5 units. The student counts 3 marks across and 30 marks up.

(a) How many marks up should the student have counted?

Answer: \_\_\_\_\_

(b) Counting 30 marks up, what  $y$ -value did the mark really land on?

Answer: \_\_\_\_\_

(c) Explain the two separate things that went wrong.

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